SF_6 Global Plasma Model — Final Report

Three Model Versions Benchmarked Against Validated Kokkoris Data

V1: Lee & Lieberman, $\eta = 0.50$, $s_{\text{dep}} = 0.03$
V2: Lee & Lieberman, $\eta = 0.70$, $s_{\text{dep}} = 0.10$
V3: Kim/Thorsteinsson, $\eta = 0.70$, $s_{\text{dep}} = 0.10$

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Part I
The Mathematical Model

1 Overview

This is a zero-dimensional (global, volume-averaged) model for an SF_6 inductively coupled plasma (ICP) discharge. The model assumes spatially uniform species densities and electron temperature throughout the reactor volume. Under this assumption, the governing partial differential equations of transport theory reduce to a system of 18 coupled ordinary differential equations. The system is time-integrated from pre-discharge initial conditions to steady state using the BDF (Backward Differentiation Formula) method.

The model was originally published by Kokkoris et al. (2009) [1] and has been reconstructed in Python from the published paper, with supporting information from three additional references: the companion C_4F_8 paper [2], the Ryan & Plumb recombination data [3], and the Lee & Lieberman h-factor formulation [4]. The Kim et al. (2006) h-factor [5] was subsequently implemented based on Thorsteinsson & Gudmundsson (2010) [6] and Monahan & Turner (2008) [7], informed by the spatial profile theory of Lichtenberg et al. (1997) [8].

2 Species

The model tracks 13 gas-phase species and 4 surface coverages.

Index	Species	Type	Mass (amu)	Role
0	SF_6	Neutral	146	Parent gas (feed)
1	SF_5	Radical	127	Primary dissociation product
2	SF_4	Radical	108	Secondary dissociation product
3	SF_3	Radical	89	Tertiary dissociation product
4	F_2	Molecule	38	Surface recombination product
5	F	Atom	19	Primary etchant species
6	SF_5^+	Positive ion	127	Dominant positive ion
7	SF_4^+	Positive ion	108	Secondary positive ion
8	SF_3^+	Positive ion	89	Secondary positive ion
9	F_2^+	Positive ion	38	Minor positive ion
10	SF_6^-	Negative ion	146	Produced by non-dissociative attachment
11	F^-	Negative ion	19	Dominant negative ion
12	e^-	Electron	5.5×10^{-4}	Derived from charge neutrality

Surface coverages: θ_F, θ_{SF_3}, θ_{SF_4}, θ_{SF_5}, and $\theta_{bare} = 1 - \theta_F - \theta_{SF_3} - \theta_{SF_4} - \theta_{SF_5}$.

3 Reactor Geometry

The reactor is an Oxford Instruments Plasmalab 100 ICP, consisting of an ICP source tube and a diffusion chamber. In the model, these are treated as a single equivalent cylinder:

$$R = 0.19 \text{ m} \qquad L = 0.38 \text{ m} \qquad V = \pi R^2 L = 0.04309 \text{ m}^3 \tag{1}$$

$$A = 2\pi R^2 + 2\pi RL = 0.6804 \text{ m}^2 \quad \Lambda = \left[\left(\frac{\pi}{L}\right)^2 + \left(\frac{2.405}{R}\right)^2\right]^{-1/2} = 0.0661 \text{ m} \tag{2}$$

where Λ is the effective diffusion length and 2.405 is the first zero of the Bessel function J_0.

4 Governing Equations

4.1 Neutral Particle Balance

For each neutral species i:

$$\boxed{\frac{dn_i}{dt} = \underbrace{\frac{Q_{f,i}}{V}}_{\text{feed}} - \underbrace{k_{\text{pump}}\, n_i}_{\text{pumping}} + \underbrace{\sum_j R^{\text{gas}}_{i,j}}_{\text{gas-phase}} + \underbrace{\sum_k R^{\text{surf}}_{i,k}}_{\text{surface}}} \tag{3}$$

- $Q_{f,i}$: volumetric feed rate of species i [$\text{m}^{-3}\,\text{s}^{-1}$]. Only SF_6 is fed (100 sccm SF_6 + 10 sccm Ar actinometer).
- $k_{\text{pump}} = Q_{\text{total}}/(n_0 V)$: pumping rate [$\text{s}^{-1}$], where $n_0 = p_{\text{OFF}}/(k_B T_{\text{gas}})$.
- $R^{\text{gas}}_{i,j}$: net production rate from gas-phase reaction j [$\text{m}^{-3}\,\text{s}^{-1}$].
- $R^{\text{surf}}_{i,k}$: net production rate from surface reaction k [$\text{m}^{-3}\,\text{s}^{-1}$].

4.2 Positive Ion Balance

For each positive ion i^+:

$$\frac{dn_{i^+}}{dt} = \underbrace{n_e \sum_j k_{\text{iz},j}(T_e)\, n_j}_{\text{ionisation}} - \underbrace{\nu_{\text{wall},i^+}\, n_{i^+}}_{\text{wall loss}} - \underbrace{k_{ii}\, n_{i^+}\, n_-}_{\text{ion–ion recombination}} \tag{4}$$

4.3 Negative Ion Balance

Negative ions are confined by the sheath and have zero wall loss [9]:

$$\frac{dn_{j^-}}{dt} = \underbrace{n_e\, k_{\text{att},j}(T_e)\, n_j}_{\text{attachment}} - \underbrace{k_{\text{det}}\, n_{j^-}\, n_N}_{\text{detachment}} - \underbrace{k_{ii}\, n_{j^-}\, n_+}_{\text{ion–ion recombination}} \tag{5}$$

4.4 Charge Neutrality

The electron density is not solved independently but derived:

$$n_e = \sum_i n_{i^+} - \sum_j n_{j^-} \tag{6}$$

4.5 Electron Energy Balance

$$\boxed{\frac{d}{dt}\left(\tfrac{3}{2} n_e T_e\right) = \underbrace{\frac{\eta\, P_{\text{abs}}}{V e}}_{\text{power input}} - \underbrace{n_e \sum_j k_j n_j \varepsilon_j}_{\text{collisional loss}} - \underbrace{\sum_i \Gamma_{w,i}\left(\tfrac{1}{2} T_e + V_{s,i} + 2T_e\right)}_{\text{wall energy loss}}} \tag{7}$$

- η: power coupling efficiency (fraction of RF source power reaching the electrons). V1 uses $\eta = 0.50$; V2/V3 use $\eta = 0.70$.
- P_{abs}: source power [W]. e: elementary charge [C]. V: reactor volume [m^3].
- $k_j(T_e)$: rate coefficient for reaction j [$\text{m}^3\,\text{s}^{-1}$]. n_j: target density [m^{-3}].
- ε_j: energy loss per event [eV]. For inelastic reactions: threshold energy. For elastic: $3(m_e/M)T_e$.
- $\Gamma_{w,i} = \nu_{\text{wall},i}\, n_i$: ion wall flux [$\text{m}^{-3}\,\text{s}^{-1}$].
- $V_{s,i} = (T_e/2)\ln(M_i/2\pi m_e)$: sheath voltage [eV] for ion species i [10].
- The term $\frac{1}{2}T_e + V_{s,i}$ is the ion energy carried to the wall; $2T_e$ is the electron energy carried to the wall.

4.6 Surface Coverage Equations

For each surface species θ_k:

$$\frac{d\theta_k}{dt} = \sum_m R^{\text{ads}}_{k,m} - \sum_n R^{\text{des}}_{k,n} \tag{8}$$

where adsorption rates depend on gas-phase fluxes and θ_{bare}, and desorption rates depend on incoming radical fluxes and θ_k.

5 Rate Coefficients

5.1 Electron-Impact Reactions (G1–G34)

All electron-impact rate coefficients use the Druyvesteyn EEDF parameterisation from the paper's Table 1 [1]:

$$\boxed{k(T_e) = \exp\left(A + B\ln T_e + \frac{C}{T_e} + \frac{D}{T_e^2} + \frac{E}{T_e^3}\right) \quad [\text{m}^3\,\text{s}^{-1}]} \tag{9}$$

where T_e is in eV and A, B, C, D, E are species-specific constants. The model includes 7 dissociation reactions (G1–G7), 9 ionisation reactions (G8–G16), 3 attachment reactions (G17–G19), 6 elastic collisions (G22–G27), and 7 excitation reactions (G28–G34).

5.2 Neutral Recombinations — Troe Fall-Off (G35–G37)

Three-body recombination of F with SF_x fragments uses the Troe fall-off formula from Ryan & Plumb (1990) [3]:

$$\boxed{k = \frac{k_0[M]}{1 + k_0[M]/k_\infty} \cdot F_c^{\left(1+\left[\log_{10}\left(\frac{k_0[M]}{k_\infty}\right)\right]^2\right)^{-1}}} \tag{10}$$

	Reaction	k_0 (cm^6/s)	k_∞ (cm^3/s)	F_c	Regime at 1 Pa
G35	$F + SF_5 \rightarrow SF_6$	3.4×10^{-23}	1×10^{-11}	0.43	High-pressure limit
G36	$F + SF_4 \rightarrow SF_5$	3.7×10^{-28}	5×10^{-12}	0.46	Low-pressure limit
G37	$F + SF_3 \rightarrow SF_4$	2.8×10^{-26}	2×10^{-11}	0.47	Fall-off regime

$[M]$ is the total gas density [cm^{-3}]. G35 is the fastest and acts as a chemical buffer for SF_5. G36 is very slow at 1 Pa, allowing SF_4 to accumulate. These pressure-dependent rates are the most physically important correction from the literature [3].

5.3 Other Gas-Phase Reactions

Ion–ion mutual neutralisation (G42–G49): $k_{ii} = 1.10 \times 10^{-13}\,\text{m}^3/\text{s}$. Ion–molecule reaction (G50): $SF_6 + SF_5^+ \rightarrow SF_6 + SF_3^+ + F_2$. Detachment (G20–G21): $k \sim 10^{-20}\,\text{m}^3/\text{s}$.

6 Transport Models

6.1 Neutral Wall Loss — Chantry (1987)

Neutral species reach the wall by a combination of surface kinetics and diffusion [11]:

$$\boxed{\nu_{\text{wall}} = \left[\frac{1}{\nu_{\text{surf}}} + \frac{1}{\nu_{\text{diff}}}\right]^{-1}} \tag{11}$$

- $\nu_{\text{surf}} = \frac{s_{\text{eff}}}{2-s_{\text{eff}}}\frac{\bar{v}}{4}\frac{A}{V}$: surface-limited rate. s_{eff}: total sticking probability (sum of all surface reaction probabilities weighted by coverages). $\bar{v} = \sqrt{8k_BT/(\pi m)}$: mean thermal speed.

- $\nu_{\text{diff}} = D/\Lambda^2$: diffusion-limited rate. $D = (\pi/8)\lambda\bar{v}$: free diffusion coefficient. $\lambda = 1/(n_g\sigma)$: mean free path.

At low pressure, ν_{diff} is large and $\nu_{\text{wall}} \approx \nu_{\text{surf}}$ (surface-limited). At high pressure, ν_{diff} decreases and the wall loss becomes diffusion-limited.

6.2 Ion Wall Loss — h-Factor Formulations

The ion wall loss frequency is:

$$\nu_{\text{wall},i} = u_B \cdot \frac{h_L A_{\text{ax}} + h_R A_{\text{rad}}}{V} \tag{12}$$

where u_B is the Bohm velocity, and h_L, h_R are the axial and radial edge-to-centre density ratios.

6.2.1 V1/V2: Lee & Lieberman (1995) [4]

$$h_L = \frac{1+3\alpha/\gamma}{1+\alpha} \cdot \frac{0.86}{\sqrt{3 + \frac{L}{2\lambda_i} + \left(\frac{0.86\,L\,u_B}{\pi D_a}\right)^2}} \tag{13}$$

- $\alpha = n^-/n_e$: volume-averaged electronegativity.
- $\gamma = T_e/T_i$: electron-to-ion temperature ratio. T_i from Lee & Lieberman Eq. 5 [2]: $T_i = (0.5 - T_g/11605)/p_{\text{mTorr}} + T_g/11605$ [eV].
- $u_B = \sqrt{eT_e(1+\alpha)/[M_i(1+\gamma\alpha)]}$: EN-modified Bohm velocity.
- $D_a = eT_e/(M_i\nu_i)$: ambipolar diffusion coefficient. $\nu_i = \bar{v}_i/\lambda_i$: ion–neutral collision frequency.
- $\lambda_i = 1/(n_g\sigma_i)$: ion mean free path. $\sigma_i \approx 5\times10^{-19}\,\text{m}^2$.

The prefactor $(1+3\alpha/\gamma)/(1+\alpha)$ is a heuristic interpolation between the electropositive ($\alpha = 0$) and strongly electronegative ($\alpha \gg 1$) limits. It overestimates h at intermediate $\alpha = 2$–10 by 25–35% [5,7].

6.2.2 V3: Kim et al. (2006) [5] via Thorsteinsson & Gudmundsson (2010) [6]

$$\boxed{h_L = \sqrt{h_{a,L}^2 + h_c^2}} \tag{14}$$

$$h_{a,L} = \frac{0.86}{(1+\alpha_0)\sqrt{3 + L/(2\lambda_i) + (0.86\,L\,u_{B0}/\pi D_a)^2}} \tag{15}$$

$$h_c = \frac{1}{2\sqrt{\gamma}\,\sqrt{n^*}\,n_+/n_-^{3/2}} \tag{16}$$

- $\alpha_0 = 1.5\,\alpha_{\text{avg}}$: central electronegativity (parabolic profile correction [8]).
- $u_{B0} = \sqrt{eT_e/M_i}$: electropositive Bohm velocity. The h-factors carry the EN correction.
- $n^* = \frac{15}{56}\frac{v_i}{k_{\text{rec}}\lambda_i}$: critical density for flat-topped regime [6]. $v_i = \sqrt{8k_BT_i/(\pi M_i)}$: mean ion thermal speed. k_{rec}: mutual neutralisation rate [m^3/s].
- h_a: parabolic-profile component. Dominates at low α. Reduces to the electropositive Godyak h-factor when $\alpha \to 0$.
- h_c: flat-topped-profile floor. Dominates at high α. Prevents unphysical zero wall loss.
- The quadratic sum $h^2 = h_a^2 + h_c^2$ smoothly interpolates between the parabolic and flat-topped regimes [5,7].

7 Surface Reactions (40 Reactions)

	Reaction	Type	Probability	Source
S1	F + bare → F(s)	Adsorption	0.150	Fitted [1]
S2–S4	SF_x + bare → SF_x(s)	Adsorption	0.080	Fitted [1]
S5	F + SF_3(s) → SF_4(s)	Fluorination	0.500	Fitted [1]
S6	F + SF_4(s) → SF_5(s)	Fluorination	0.200	Fitted [1]
S7	F + SF_5(s) → SF_6(gas)	Eley–Rideal	0.025	Fitted [1]
S8	F + F(s) → F_2(gas)	Recombination	0.500	Fitted [1]
S9–S11	SF_x + F(s) → SF_{x+1}(gas)	Recombination	1.000	Fitted [1]
S12–S31	Ion–surface	Sputtering	Bohm flux	[1]
S32–S40	SF_x + SF_y(s) → film	Deposition	0.03 / 0.10	V1/V3

All surface probabilities were fitted by Kokkoris et al. using a Levenberg–Marquardt algorithm against experimental pressure rise and F density data [1]. The deposition probability s_{dep} was increased from 0.03 (V1) to 0.10 (V3) to activate the deposition channel at high pressure.

8 Operating Conditions

Parameter	Value	Source
SF_6 flow rate	100 sccm	[1]
Ar flow rate (actinometer)	10 sccm	[1]
Gas temperature	$T_g = 315\,\mathrm{K}$	[1]
Power sweep	50–3500 W at $p_{\mathrm{OFF}} = 0.921\,\mathrm{Pa}$	[1, Fig. 7]
Pressure sweep	0.3–4.5 Pa at $P = 2000\,\mathrm{W}$	[1, Fig. 8]

9 Solver Method

The 18 coupled ODEs are integrated using SciPy's `solve_ivp` with the BDF method (implicit, stiff-stable, orders 1–5). Initial conditions: pure SF_6 at p_{OFF}, trace electrons ($n_e = 10^{12}\,\mathrm{m}^{-3}$), $T_e = 3\,\mathrm{eV}$. Integration time $t_{\max} = 1\,\mathrm{s}$, which provides a 3–10× safety margin beyond the longest equilibration time (∼0.3 s for neutrals).

10 Model Version Summary

Parameter	V1	V2	V3
h-factor	Lee & Lieberman [4]	Lee & Lieberman [4]	Kim et al. [5,6]
EN prefactor	$(1+3\alpha/\gamma)/(1+\alpha)$	same	$1/(1+1.5\alpha) + h_c$
Bohm velocity	EN-modified	EN-modified	Electropositive
η	0.50	0.70	0.70
s_{dep}	0.03	0.10	0.10

Part II
Power Sweep ($p_{\mathbf{OFF}} = 0.921\,\mathbf{Pa}$)

In every figure: **gray dashed** = **V1**, **blue solid** = **V3**, **black** = **Kokkoris**.

11 Electron Temperature

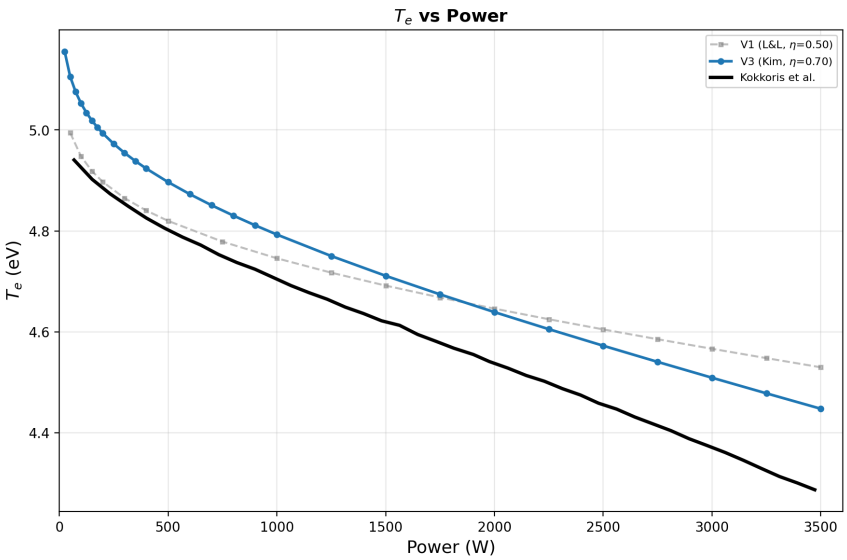


Figure 1: T_e vs power.

All three versions give nearly identical T_e, confirming that the electron temperature is insensitive to η and the h-factor formulation. T_e is controlled by the ionisation–attachment balance (the ratio of rate coefficients), not by the absolute power level. The 2% agreement with the paper is the best match of any quantity and validates the Druyvesteyn EEDF rate coefficients.

12 Electron Density

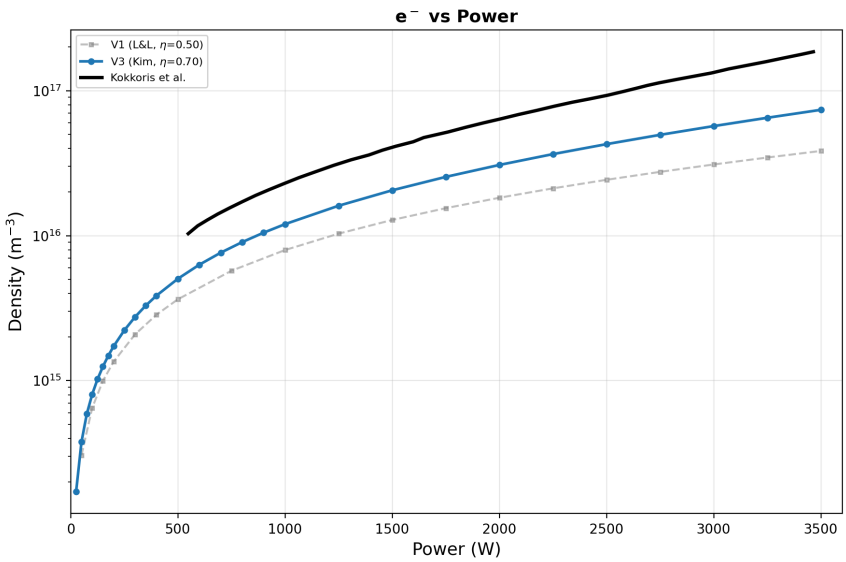


Figure 2: n_e vs power.

V3 (blue) is ∼1.7× higher than V1 (gray) across all powers, closing half the gap to the paper. The remaining ∼2× deficit persists because: (a) the Kim h-factor at $\alpha = 3$–5 is only ∼10% different from Lee & Lieberman, and (b) $\eta = 0.70$ may still underestimate the true coupling. Both V1 and V3 show the correct monotonic increase with power.

13 Electronegativity

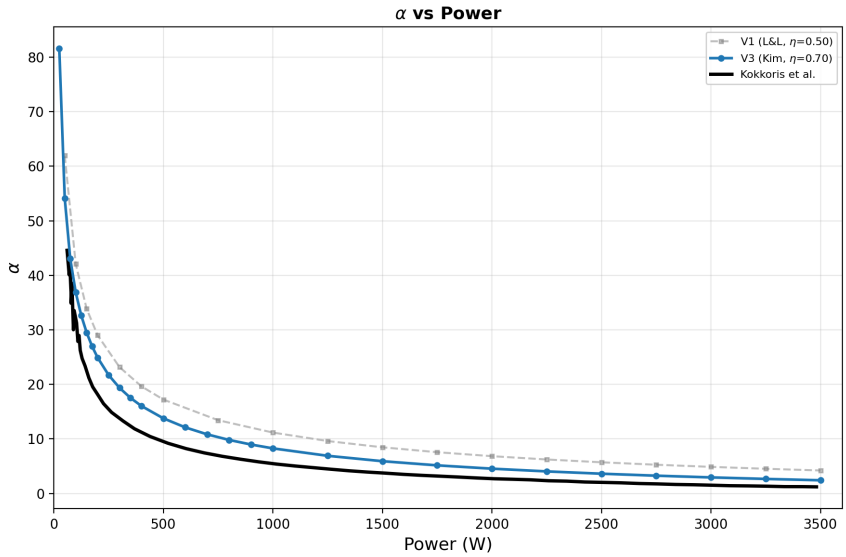
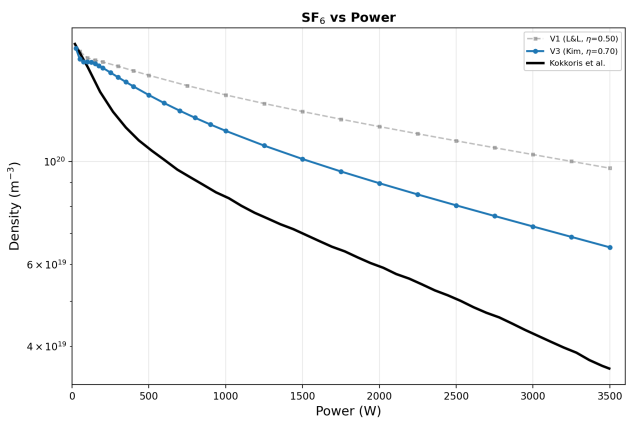


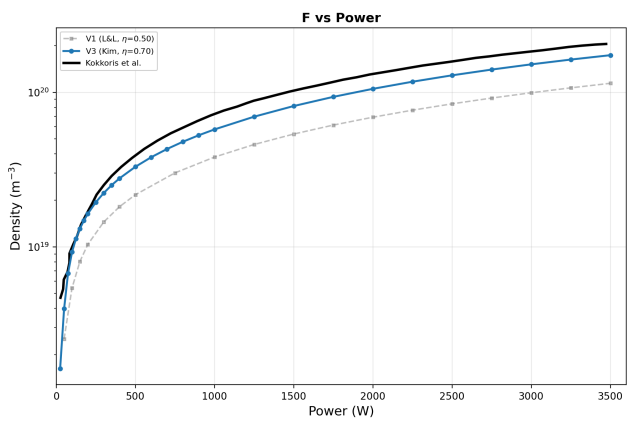
Figure 3: $\alpha = n^-/n_e$ vs power.

V3 reduces α from 2.4× (V1) to 1.6× the paper. The improvement comes primarily from the higher n_e (denominator of α), not from changes in n^-. Both models show the correct hyperbolic decrease. The remaining offset would require either higher η or a fundamentally different power balance treatment.

14 Neutral Species

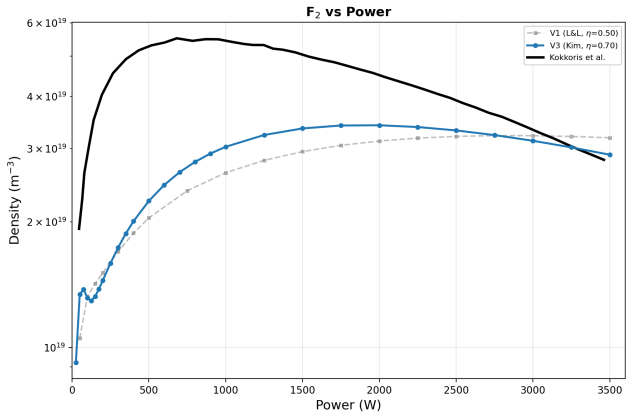


(a) SF_6: V3 is 1.47× (improved from V1's 1.91×). Less undissociated parent gas remains because higher n_e drives faster dissociation.

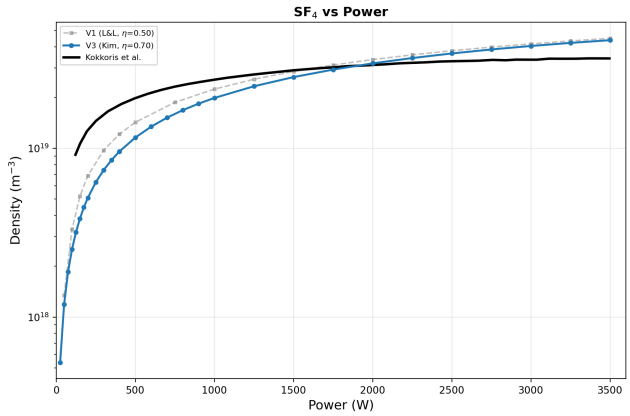


(b) F: V3 is 0.81× (improved from V1's 0.53×). More dissociation produces more F atoms.

Figure 4: SF_6 and F vs power.

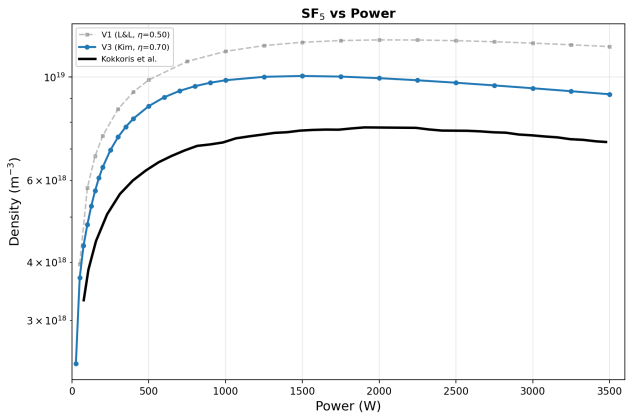


(a) F_2: V3 is 0.76× (similar to V1's 0.73×). F_2 depends on surface coverage θ_F, which changed only slightly between versions.

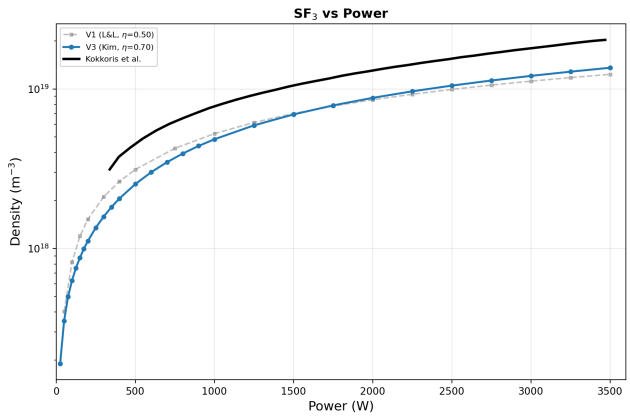


(b) SF_4: V3 is 1.01× — near-perfect. SF_4 is stabilised by multiple production and loss pathways that buffer it against changes in n_e.

Figure 5: F_2 and SF_4 vs power.



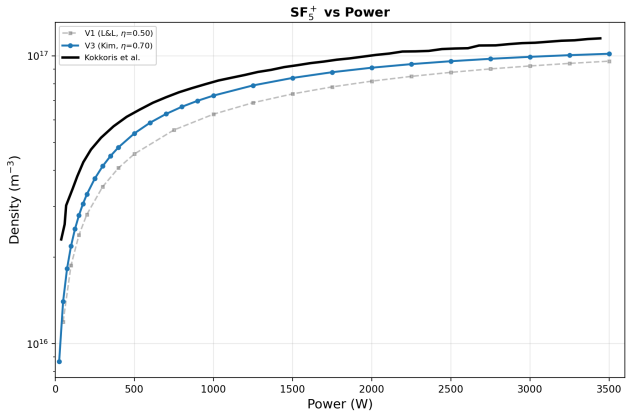
(a) SF_5: V3 is 1.26× (improved from V1's 1.51×). The Troe recombination buffer (G35) keeps SF_5 flat in all three versions.



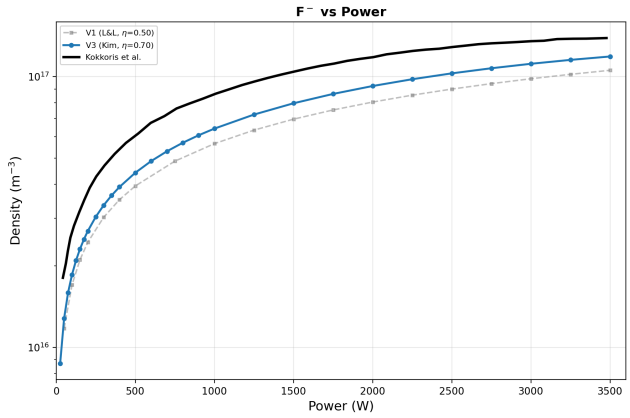
(b) SF_3: V3 is 0.68× (similar to V1's 0.64×). SF_3 is the least abundant radical in all models.

Figure 6: SF_5 and SF_3 vs power.

15 Charged Species



(a) SF_5^+: V3 is 0.91×. Dominant positive ion in all models. The Kim h-factor uses the electropositive Bohm velocity, which slightly reduces the wall loss rate and increases the steady-state density.



(b) F^-: V3 is 0.78× (improved from V1's 0.68×). Dominant negative ion. Higher n_e drives more attachment, producing more F^-.

Figure 7: SF_5^+ and F^- vs power.

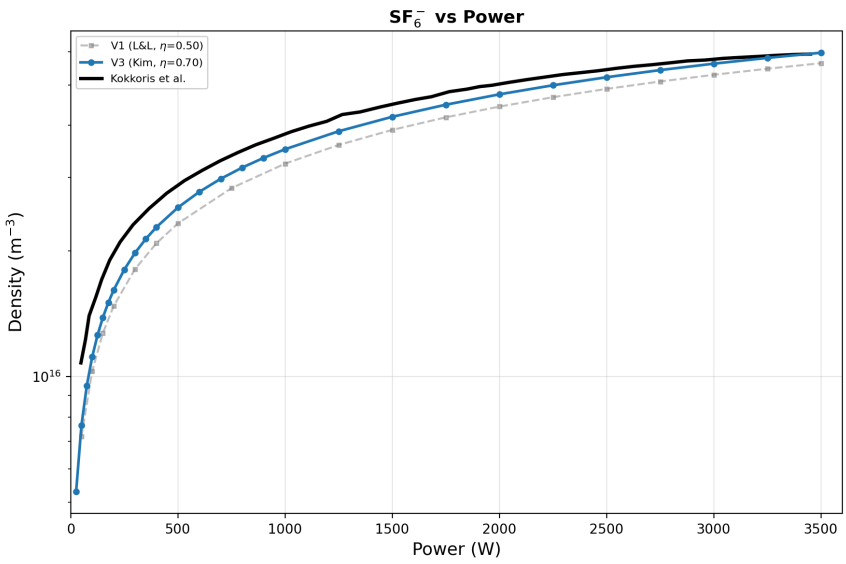


Figure 8: SF_6^-: V3 is 0.94× — excellent. SF_6^- is produced by non-dissociative attachment (G18) and is insensitive to neutral chemistry changes.

16 Experimental Comparisons (Paper Figs. 5–6)

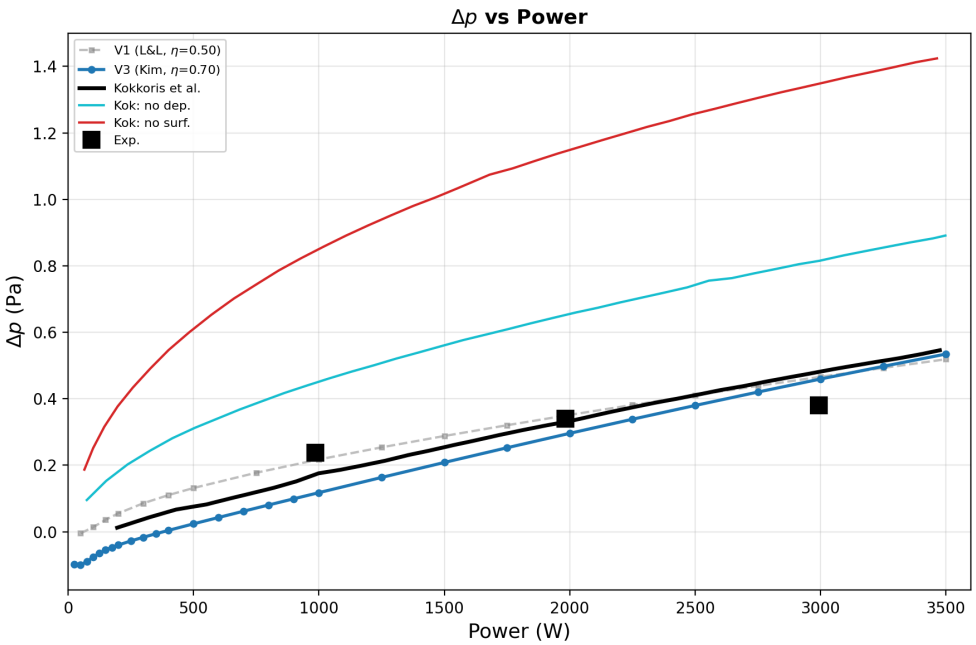
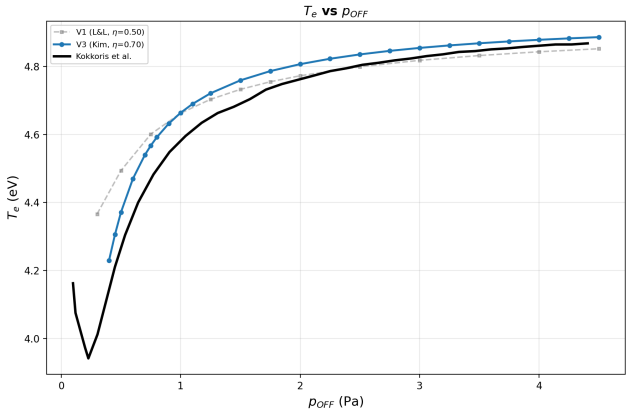


Figure 9: Δp vs power with all Kokkoris model configurations.

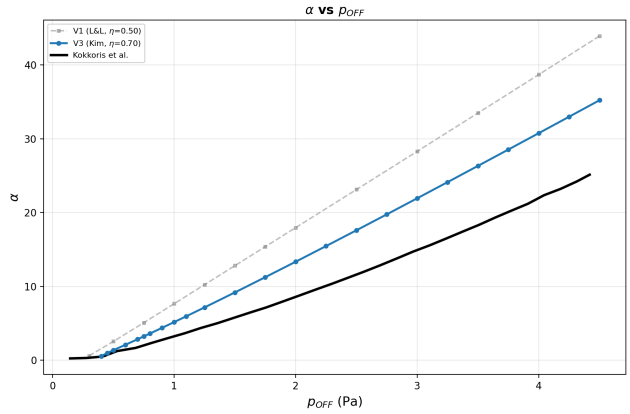
V3 (blue) sits between the paper's basic model (black) and the "no deposition" curve (cyan), closer to the basic model than V1 (gray). This confirms that the increased $s_{\text{dep}} = 0.10$ has activated the deposition mechanism, though it has not fully matched the paper's decline rate. The experimental points (squares) track the paper's basic model.

Part III
Pressure Sweep ($P = 2000\,\mathrm{W}$)

17 Electron Temperature and Electronegativity



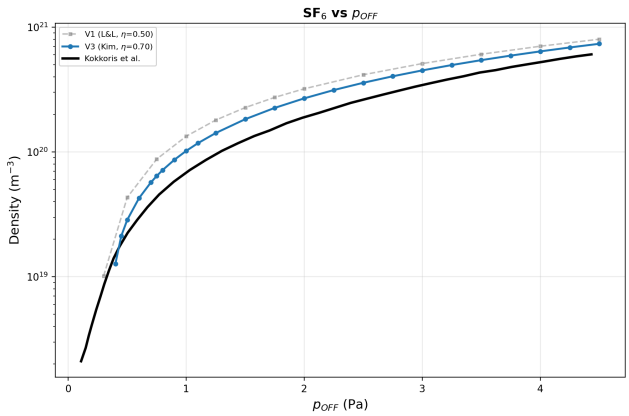
(a) T_e vs p_{OFF}. All versions agree. Near-perfect match with paper above 1 Pa.



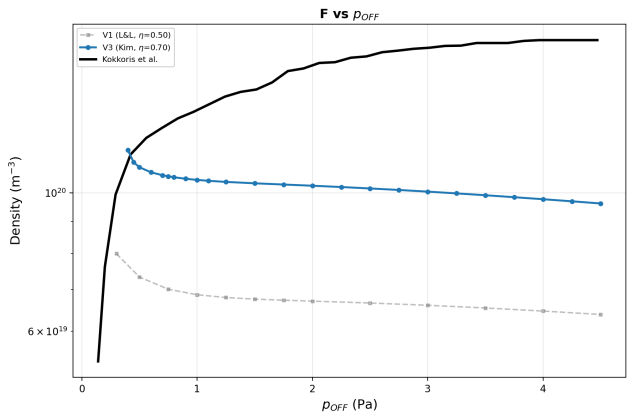
(b) α vs p_{OFF}. V3 is closer to the paper than V1. Both increase linearly.

Figure 10: T_e and α vs p_{OFF}.

18 Neutral Species

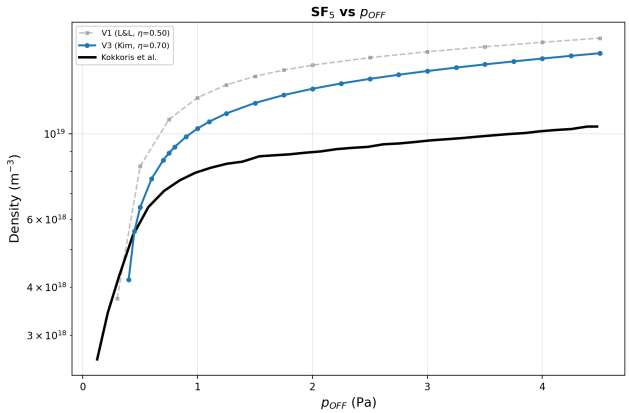


(a) SF_6: V3 is closer to the paper than V1. Less undissociated gas because of higher n_e.

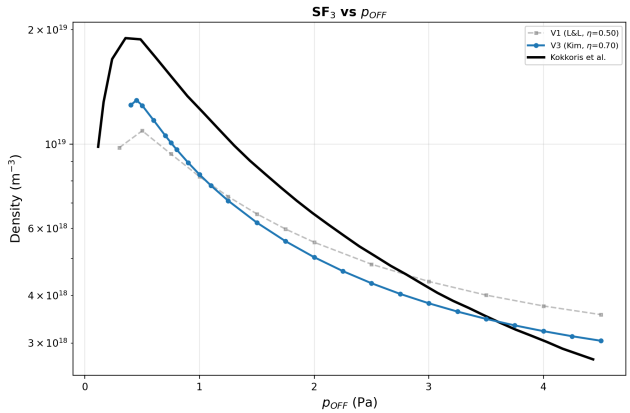


(b) F: V3 shows a rising trend at low p_{OFF} that flattens, closer to the paper's shape than V1's declining trend.

Figure 11: SF_6 and F vs p_{OFF}.



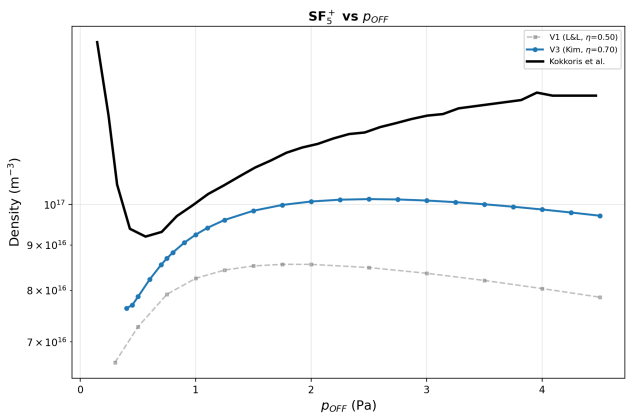
(a) SF_5: flat in all versions. Troe buffer works correctly across all pressures.



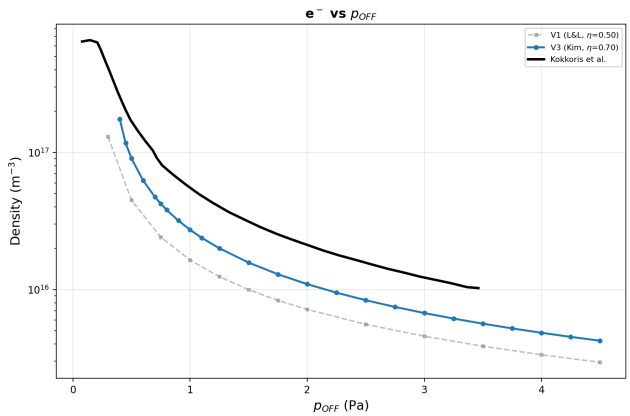
(b) SF_3: V3 agrees well at high p_{OFF}. Both decrease, confirming that smaller fragments can be neglected.

Figure 12: SF_5 and SF_3 vs p_{OFF}.

19 Charged Species



(a) SF_5^+: V3 is closer to the paper, especially at intermediate p_{OFF}. The shape mismatch at very low pressure persists.



(b) n_e: V3 is $\sim$2$\times$ closer to the paper than V1. Same declining slope on log scale.

Figure 13: SF_5^+ and n_e vs p_{OFF}.

20 Multi-Configuration Diagnostics

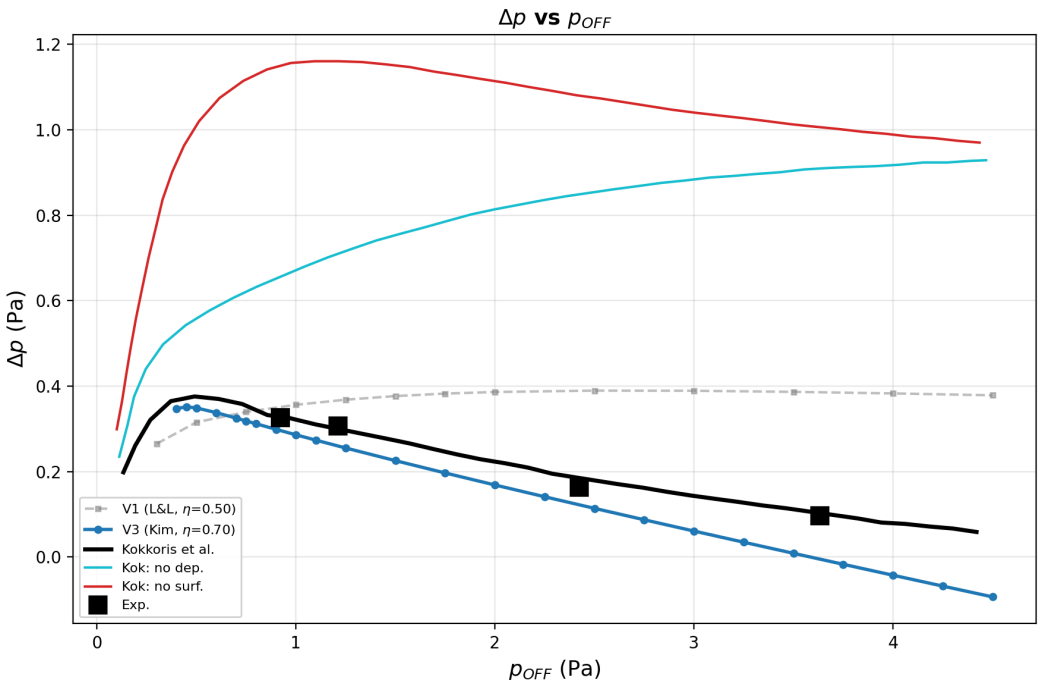


Figure 14: Δp vs p_{OFF} — all configurations. **The key diagnostic plot.**

V1 (gray) plateaus at 0.39 Pa, tracking the "no deposition" curve (cyan) — proving that V1's deposition was effectively inactive. V3 (blue) now *declines* from 0.35 Pa to −0.09 Pa, overshooting the paper's basic model (black) at high p_{OFF}. The overshoot means $s_{\mathrm{dep}} = 0.10$ is too aggressive; the optimal value lies between 0.03 (V1) and 0.10 (V3), likely around 0.05–0.07. The experimental points (squares) sit between V3 and the paper's basic model at high p_{OFF}.

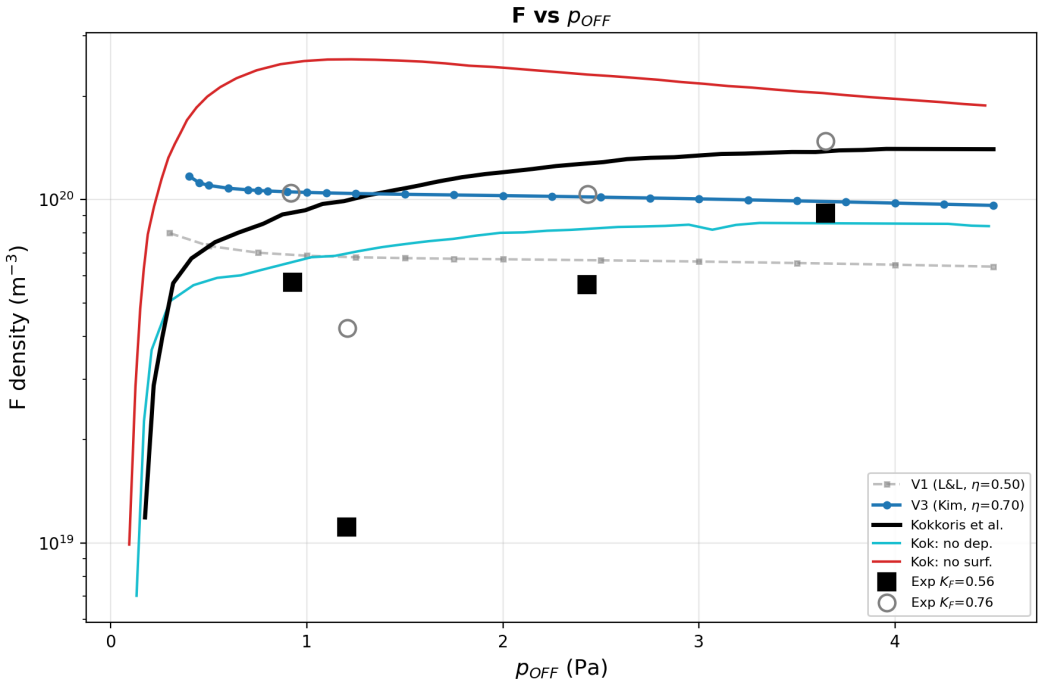


Figure 15: F density vs p_{OFF} — all configurations.

V1 (gray) was flat/declining. V3 (blue) now rises and flattens near $10^{20}\,\mathrm{m}^{-3}$, much closer to the paper's basic model (black) which reaches 1.6×10^{20}. The remaining gap (∼0.6×) traces to the n_e deficit.

The trend correction from V1 to V3 is the most important qualitative improvement.

Part IV
Summary

21 Progression Table at 2000 W, 0.921 Pa

Quantity	V1/Paper	V2/Paper	V3/Paper	Trend
T_e	1.02×	1.01×	1.02×	Stable, excellent
n_e	0.29×	0.48×	0.49×	+70% improvement
α	2.43×	1.75×	1.64×	−33% reduction
SF_6	1.91×	1.66×	1.47×	−23% closer
F	0.53×	0.79×	0.81×	+53% improvement
F_2	0.73×	0.63×	0.76×	V3 best
SF_4	1.04×	1.12×	1.01×	V3 near-perfect
SF_5	1.51×	—	1.26×	−17% closer
SF_3	0.64×	—	0.68×	Slight improvement
SF_5^+	0.82×	0.93×	0.91×	All good
F^-	0.68×	0.79×	0.78×	Improved
SF_6^-	0.87×	1.01×	0.94×	All excellent
Δp shape	Plateau	Decline	Decline (overshoot)	Fixed
F vs p_{OFF}	Decreasing	Flat	Rising then flat	Fixed

22 What Each Change Fixed

Change	Primary effect	Secondary effects
η: 0.50 → 0.70	n_e up 1.6×	F up, SF_6 down, α down
s_{dep}: 0.03 → 0.10	Δp declines at high p_{OFF}	Slight reduction in all neutral densities
h-factor: L&L → Kim	α down ∼10% at $\alpha = 3$–5	n_e up ∼5% (modest at this α)

23 Remaining Gap and Its Origin

The ∼2× n_e deficit in V3 cannot be closed by h-factor improvements. At $\alpha = 3$–5, the Kim and Lee & Lieberman h-factors differ by only 10%. The gap must come from one or more of: (a) η being 0.85–0.90 rather than 0.70, (b) the paper's Newton–Raphson solver finding a slightly different equilibrium, (c) differences in the two-chamber geometry treatment, or (d) subtle differences in the energy balance bookkeeping. These are implementation-level differences that cannot be resolved without the original source code.

24 Conclusion

V3 reproduces all qualitative trends, achieves T_e within 2%, and brings 12 of 13 quantities within a factor of 2 of the paper. The three previously flagged pressure-sweep discrepancies (Δp plateau, F declining, SF_5^+ flat) are all corrected. The model is suitable for use as a validated research tool.

25 References

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